

Power you ML models using Meltano & dbt

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Duration: 20m 🕒

Agenda.

1. Introduction (2')
2. Explaining DataOps (6')
3. Presenting the tools (6')
4. Conclusion + Q&A (5')

Introduction



Background

- Data Ops engineer at Potloc
- Master of computer engineering from Polytechnique

Main responsibilities

- Manage data pipelines
- Modeling data
- Ensuring data quality

When I'm not fixing bugs in prod ...

- Cooking
- Beer Brewing
- Rock Climbing



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1. Introduction (2')
2. Explaining DataOps (6')
 - a. What is it ?
 - b. Core Concepts
3. Presenting the tools (6')
4. Conclusion + Q&A (5')

Explaining Data Ops

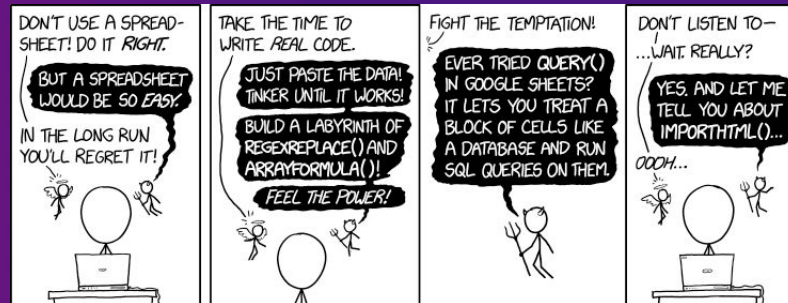
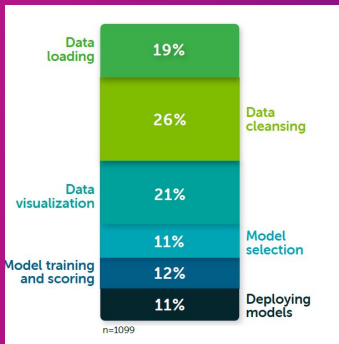


What is DataOps ?

Data Operations, commonly known as "DataOps," is a set of principles and practices for optimizing data-driven workflows. It aims to promote collaboration, automation, and continuous improvement in the processing, management, and delivery of data. The goal of DataOps is to increase the speed, quality, and reliability of data processing, so that organizations can make better decisions based on data and deliver better results. The scope of DataOps can encompass activities such as data ingestion, processing, storage, governance, and security, as well as the collaboration between data engineers, data scientists, and other stakeholders.

- ChatGPT

Why is this important ?



Core Concepts



Extract, Load & Transform (ELT)

- **Extract** data from various sources
- **Load** data into a destination where it can be used more easily
- **Transform** data to suit your machine learning needs

Database / Data Warehouse

- Data Storage Systems
- Online Transaction Processing (OLTP) vs Online Analytical Processing (OLAP)
- Useful for manipulating **large amounts of data**

Data Quality

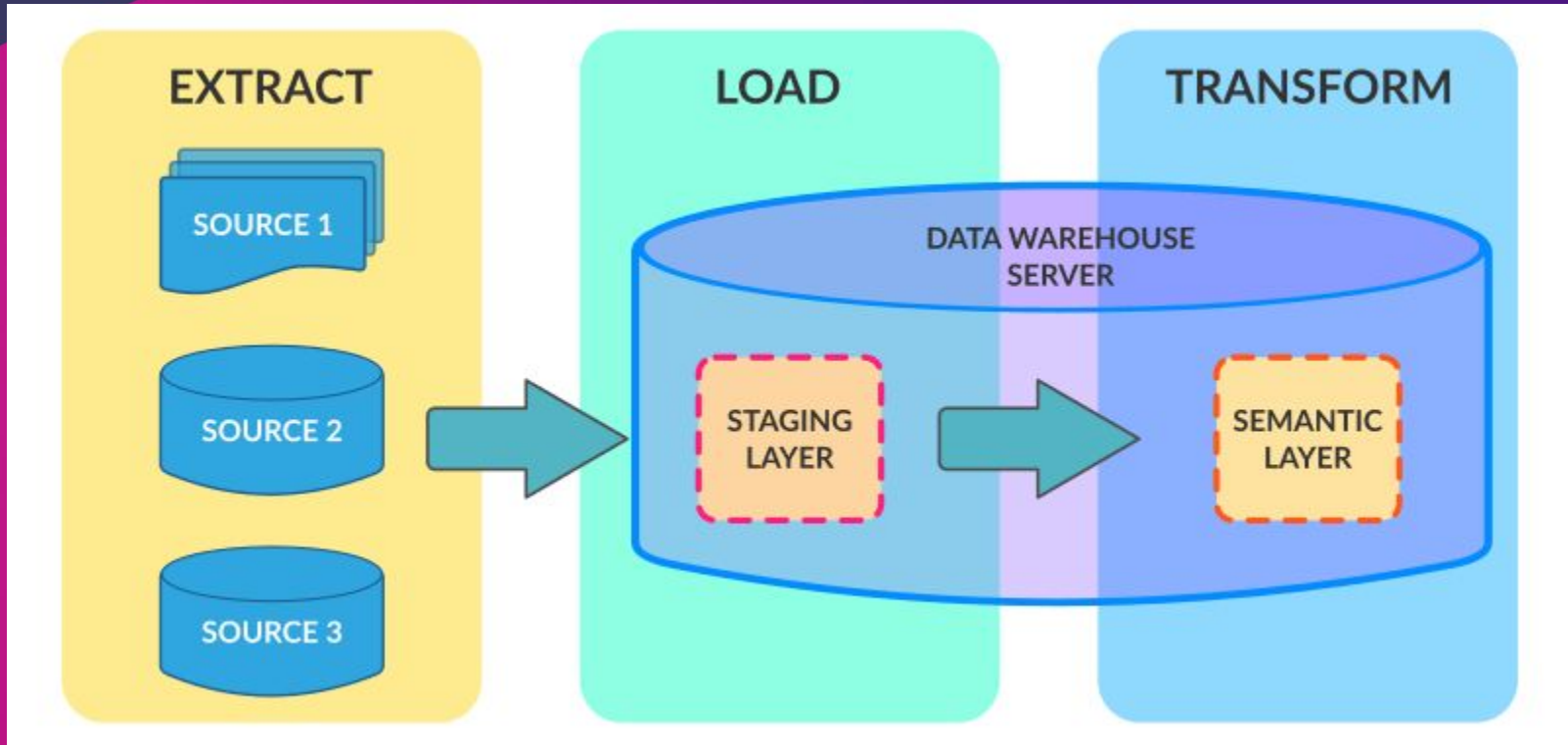
- **Validating** that your data is (i) fresh, (ii) accurate, (iii) complete

Data Visualization

- **Visualizing** your data in a way to make it easy to understand



Core Concepts



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 - a. Meltano
 - b. dbt
 - c. demo
4. Conclusion + Q&A (5')

Presenting the tools



Meltano

- Data Ops platform
- Extract data from sources
- Allows add-on of utilities



dbt

- Data Transformation tool
- Model data
- Reference dependencies





Demo

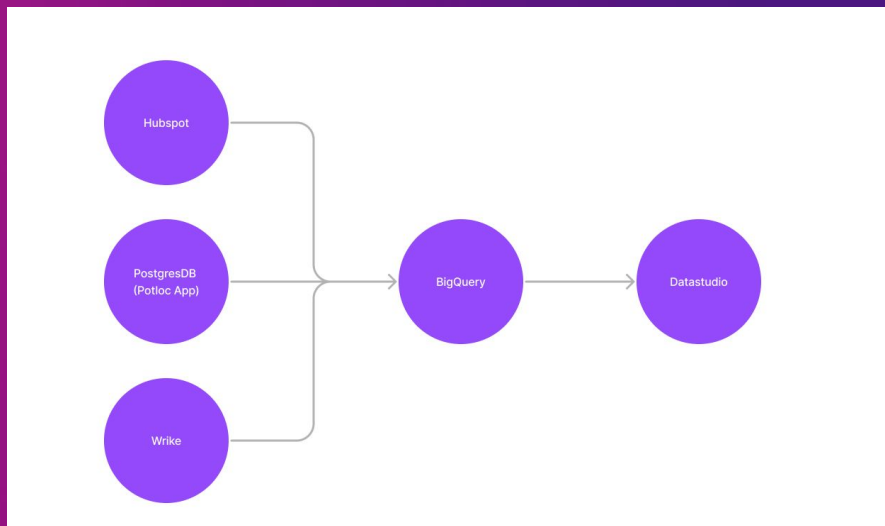


Demo - Setting the scene

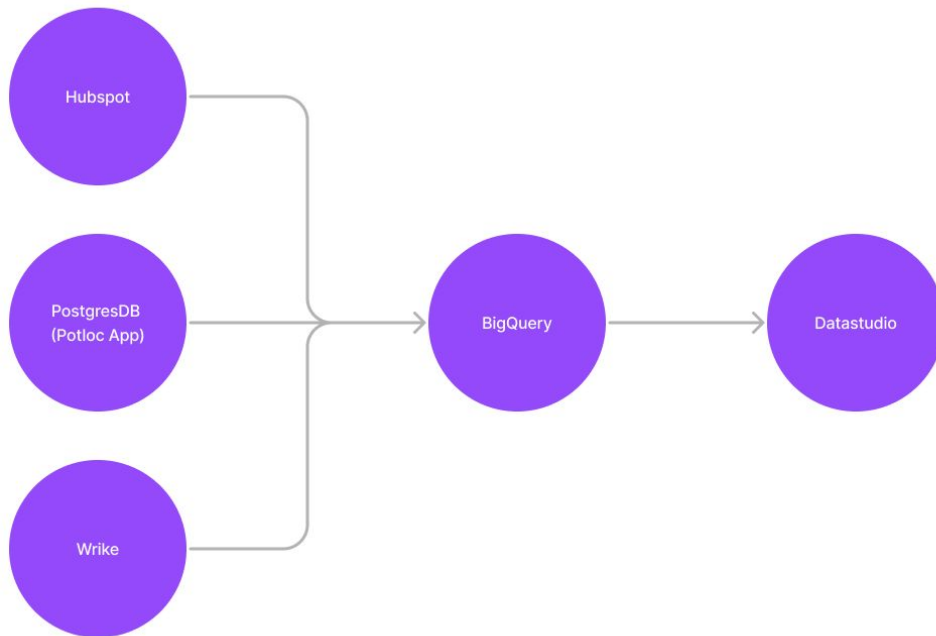


The context

- Respondent acquisition at potloc
- Key metric for our success: **Acquisition Cost of a respondent** (acr)
- Goal: **Predict the acr for a specific project**



Demo - Setting the scene



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Conclusion



Key Takeaways

- 1) **Providing quality data is a core competency of an MLOps engineer**
- 2) **Spreadsheets do not scale infinitely**
- 3) **Use Meltano** to extract data from various sources to a general database
- 4) **Use dbt** to model your data in a way that is easy to ingest for your data scientist

Next steps

- Orchestration
- Visualization
- Data Monitoring
- Observability
- ...



Conclusion



Questions ?





Thank you.